

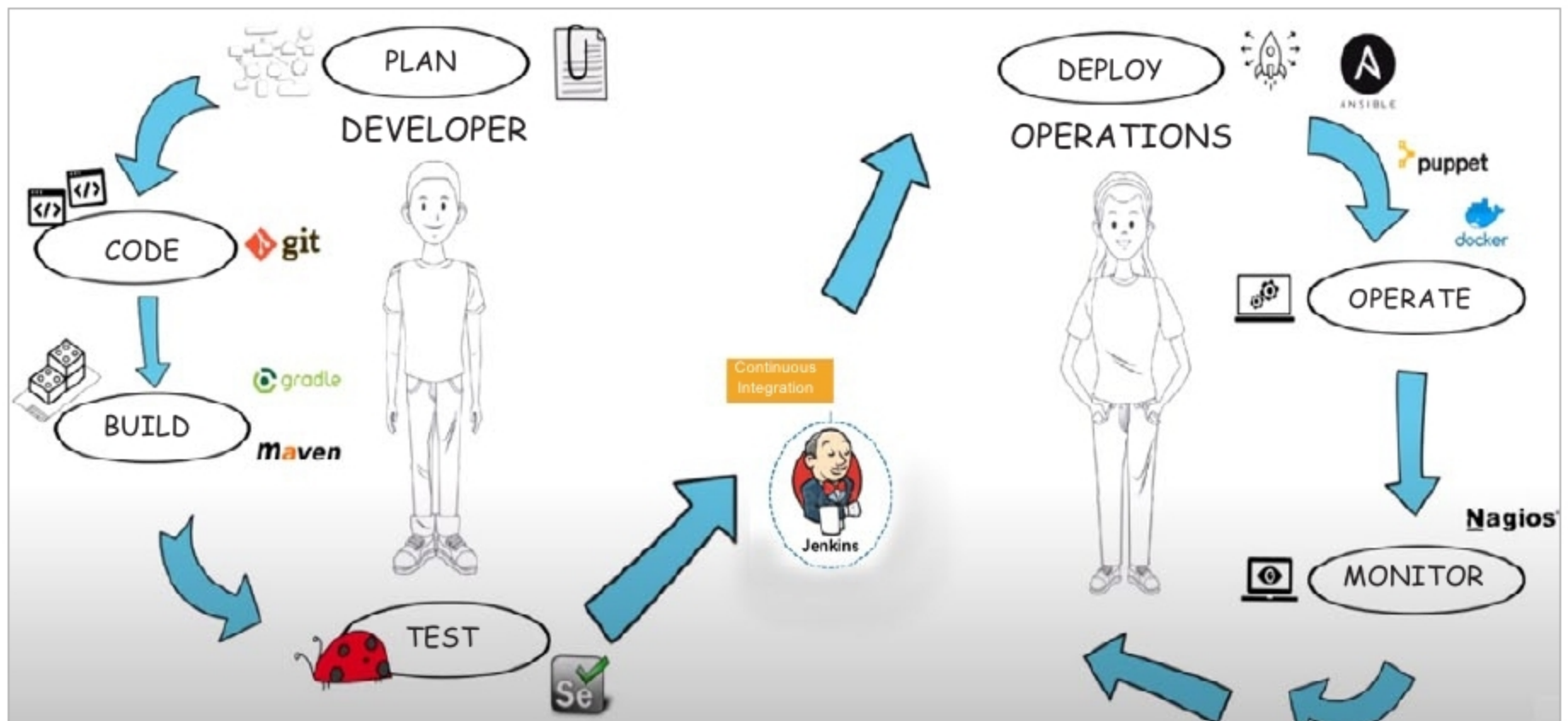


Figure 2: Categories of popular DevOps tools

reflected using the Jenkins tool.

4. Continuous deployment:

The deployed applications are consistently improved by collecting end user experiences from the obtained results of the software. Popular tools that are used for the deployment stage are Ansible, Puppet and Docker.

5. Continuous monitoring:

This is the stage when the performance of the deployed product is monitored. In this phase, the production server is continuously monitored for all kinds of activities that happen on it, and the log files are generated. Nagios is a popular tool for this phase.

Careful implementation of these five phases of the DevOps life cycle helps an IT organisation to achieve its business objectives. Various tools that contribute to the DevOps life cycle are highlighted in Figure 2.

The pros and cons of adopting DevOps

The pros and cons that should be considered when adopting DevOps are listed here.

The pros

- It facilitates faster code deployment into production.
- Ensures the release of quality products.
- It helps in reducing the chances of software failures.
- Enables continuous delivery of software with stable features.
- Helps in identifying bugs at an early stage of development due to better communication between the development and operations teams.
- Implementation of DevOps also increases the organisation's opportunities for innovation and research.

The cons

- It takes time for users to adjust to the frequent changes that are made.
- Lack of a separate plan for consistent upgradation may lead to severe shortfalls with respect to security.
- IT companies need to hire experts who have an in-depth understanding of development, testing, monitoring and deployment tools, and that can be a challenge.

With technology evolving very rapidly, IT companies that do not adopt DevOps may find it difficult to compete in the market. The adoption of the DevOps model is inevitable, because it bridges the gap between the developer and operations teams. **END** 🐧

References

These links are in addition to those provided in the body of the article itself.

- [1] <https://devops.com/>
- [2] <https://www.edureka.co/blog/devops-lifecycle/>
- [3] <https://intellipaat.com/blog/tutorial/devops-tutorial/>
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- [5] A. Bhardwaj and C. Rama Krishna, 'A Container-based Technique to Improve Virtual Machine Migration in Cloud Computing', IETE Journal of Research, pp. 1-16, 2019.

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